

memory error: VQE simulation of H2O

<https://github.com/qiskit-community/qiskit-aqua/issues/1108>

ROW 38

memory error: VQE simulation of H2O #1108



Closed



ares201005 opened on Jul 9, 2020

...

Information

- Qiskit Aqua version: 0.7.3
- Python version: 3.7
- Operating system: Linux

What is the current behavior?

The VQE simulation of H2o molecule results in memory error:

```
.../site-packages/qiskit/aqua/operators/primitive_ops/pauli_op.py
../site-packages/qiskit/quantum_info/operators/pauli.py
.../site-packages/scipy/sparse/compressed.py
.../site-packages/scipy/sparse/base.py
```

MemoryError: Unable to allocate 4.00 GiB for an array with shape (16384, 16384) and data type complex128

However, my computer has enough memory, my other script requesting much larger memory still works fine.

Steps to reproduce the problem

using H2O molecule, VQE with UCCSD ansatz, L_BFGS_B optimizer,

What is the expected behavior?

With earlier version of qiskit-aqua (0.6.5) VQE simulation of the H2O molecule does not have such a problem.

Suggested solutions

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants



woodsp-ibm on Jul 9, 2020

Member

...

Are you using statevector_simulator? If so this is most likely the same issue as [#1027](#) and in which case might I suggest you use the Aer qasm_simulator and set `use_custom` to True on VQE (see <https://qiskit.org/documentation/stubs/qiskit.aqua.algorithms.VQE.html> for docs on this). This will use the custom Aer snapshot instruction to compute the expectation (uses AerPauliExpectation object inside VQE to do this - though you can pass this in explicitly to VQE). Overall this mode is more efficient too than running with statevector and like statevector has the ideal outcome without shot noise.



ares201005 on Jul 10, 2020

Author

...

Yes, it is a statevector_simulator. Thank you for the hint, it indeed solves the problem.



adekusar-dri closed this as **completed** on Jul 14, 2020